

MD HASHIM

AI/ML Research Engineer | Quantum Machine Learning · Healthcare & Life Sciences AI
Bengaluru, India | +91-9870216640 | md.hashim127@gmail.com | linkedin.com/in/hashim--hassan | github.com/hashim-hassan

PROFESSIONAL SUMMARY

Senior AI/ML engineer with 5+ years delivering production- and research-grade AI across pharmaceutical R&D — spanning medical imaging, generative & agentic AI, and hybrid quantum-classical molecular simulation. Rare combination of novel medical-image segmentation (clinical MRI and whole-slide digital pathology, with uncertainty quantification, weakly-supervised learning, and conditional architectures), foundation-model & LLM/agentic systems (RAG, Llama, Mistral, MCP), and quantum molecular simulation (VQE, ADAPT-VQE, DMET-VQE). Recently delivered an AI pipeline for CNS drug-delivery simulation (Dice 0.951 across 42 imaging sites and 3 MRI vendors) and gigapixel pathology segmentation (Dice 0.97). Focused on foundation models and multimodal AI for drug discovery and neuroscience-driven chronic disease, and on translating complex scientific data into validated, high-impact systems. Targeting senior Research Scientist / Applied Scientist / Research Engineer roles advancing AI for science and human health.

TECHNICAL SKILLS

AI/ML Methods: Deep learning, ensemble learning (XGBoost, Random Forest, Gradient Boosting), feature engineering, explainable AI (SHAP), uncertainty quantification & model calibration, weakly-supervised & partial-label learning, transfer learning, benchmarking & error analysis, statistical validation

Deep Learning & Medical Imaging: PyTorch, TensorFlow, MONAI, CNNs, U-Net (FiLM / conditional), Vision Transformers (ViT), nnU-Net, DeepLabV3, EfficientNet, 2D/3D medical image analysis & segmentation, clinical MRI (multi-vendor), whole-slide imaging / digital pathology, domain generalization, test-time augmentation, 3D mesh generation, object detection & tracking (YOLO, DeepSORT), re-identification, OpenCV, scikit-image

Generative & Agentic AI, LLMs & NLP: Large language models (Llama, Mistral, BERT), transformers, Retrieval-Augmented Generation (RAG), agentic AI (tool use, MCP), LangChain, vector databases, prompt engineering, foundation-model fine-tuning

Quantum Computing: Qiskit, PennyLane, Cirq, D-Wave Ocean; VQE, ADAPT-VQE, DMET-VQE, UCCSD, QAOA, QPE; hybrid quantum-classical workflows, quantum circuit design, transpilation

Computational Chemistry: RDKit, PaDEL-Py, COSMO, DFT, CCSD, MP2; molecular descriptors; DMET fragmentation; basis sets (STO-3G, 6-31G, 6-311G, 6-31+G(d,p))

MLOps & Infrastructure: AWS, Docker, Kubernetes, HPC (SLURM, Apptainer), CI/CD, MLflow, Weights & Biases, TensorBoard, mixed-precision (AMP), model monitoring, automated retraining, Streamlit

Languages & Tools: Python, SQL, Bash, NumPy, Pandas, CUDA, cuDNN, Bio-Formats / OME-XML, Git, git-annex

Domain Knowledge: Drug discovery & molecular property prediction, CNS drug-delivery modeling, quantitative pathology & immunofluorescence, CFD (Ansys Fluent, OpenFOAM), biomedical & neuroimaging data (BIDS), pharmaceutical R&D

PROFESSIONAL EXPERIENCE

Machine Learning Engineer

Eli Lilly · Bengaluru, India

Dec 2024 – Present

Brain Tissue Segmentation for Neuroscience Drug Programs — PyTorch, Depth-Conditioned U-Net (FiLM), EfficientNet-B4, HPC (SLURM/Apptainer)

- Designed a novel Depth-Conditioned U-Net with Feature-wise Linear Modulation (FiLM) injecting tissue-depth embeddings into encoder and decoder, replacing per-depth ensembles with a single depth-aware model for 5-class gigapixel whole-slide immunofluorescence segmentation (brain stem, cerebellum, frontal cortex, hippocampus, striatum), reaching Dice 0.97 / IoU 0.92 across NHP and mouse drug programs.
- Combined a multi-seed EfficientNet-B4 ensemble with compound Dice+BCE loss, dynamic class weighting, and CopyPaste augmentation, plus a masked-loss framework that zeroes gradients on unannotated classes, enabling weakly-supervised training on partial pathologist labels and stabilizing extreme minority-class performance.
- Built an end-to-end pipeline from raw microscopy and pathologist annotations (metadata extraction, mask rasterization, group-aware splitting) through full-resolution tiled inference with test-time augmentation to HALO-compatible XML export for pathologist review in Indica Labs.
- Containerized the workflow on HPC (SLURM, Apptainer) for reproducible scanner-to-HALO execution, replacing days-long manual annotation with minutes-per-study batch inference for high-throughput quantitative pathology in drug-efficacy analysis.

- **Drug-discovery ML:** Built ensemble models (XGBoost, Random Forest, Gradient Boosting) for molecular screening at >85% accuracy in identifying viable candidates; engineered molecular-descriptor pipelines (RDKit, PaDEL-Py) and a SHAP-based digital twin surfacing the chemical features that drive medicinal-chemist prioritization.
- **GenAI & agentic assistant (RAG):** Architected an enterprise Retrieval-Augmented Generation assistant (Llama, Mistral, LangChain, MCP tool use) serving product knowledge for Mounjaro, Zepbound, and Ebglyss via vector-retrieval and SQL knowledge pipelines.
- **Hybrid predictive modeling:** Designed a framework coupling COSMO thermodynamic simulation with ML to improve reliability of molecular-property prediction.
- **MLOps deployment:** Deployed models on AWS Kubernetes with CI/CD and automated retraining, enabling scalable production rollout.

Machine Learning Engineer, Computer Vision

Tata Consultancy Services • Bengaluru, India

Apr 2021 – Nov 2024

Official TCS designation: Systems Engineer

- **Multi-camera tracking:** Built a multi-camera, multi-object tracking system (YOLO, DeepSORT, Re-Identification) for persistent tracking across distributed surveillance environments, resolving ID-switching with a ReID vector and ID-correction model.
- **Real-time video analytics:** Developed detection, tracking, and behavioral-analysis pipelines on live video streams using reusable PyTorch / TensorFlow components.
- **Healthcare analytics:** Analyzed biomechanical posture data for chronic back pain (K-Means, DBSCAN) to support health-risk profiling and premium planning.
- **DevOps automation:** Automated data preprocessing, model training, and deployment workflows with Docker and CI/CD.

KEY PROJECTS

AI Digital Twin of the Human CNS for Drug-Delivery Simulation

MONAI • PyTorch • 3D segmentation • uncertainty quantification • CFD (Ansys Fluent / OpenFOAM) • AI segmentation-and-meshing engine powering CFD simulation of intrathecal drug delivery, efficacy, and toxicity for CNS therapeutics.

- Delivered an end-to-end 3D segmentation pipeline reaching Dice 0.951 on spinal-cord segmentation across 42 sites and 3 MRI vendors (GE, Philips, Siemens), surpassing the Spinal Cord Toolbox baseline; held inter-site variation below 5% via normalization, site-stratified validation, and spatial augmentation.
- Engineered a topology-aware training framework (custom Soft-cIDice + DiceCE loss) yielding 100% watertight, manifold meshes, with a CFD-ready export chain converting segmentation masks to STL geometries for direct Ansys Fluent / OpenFOAM ingestion.
- Added Monte Carlo Dropout uncertainty quantification with calibrated confidence (ECE = 0.005, ~10x below clinical threshold), enabling automated quality-control flagging for expert review.
- Generalized across spinal cord (binary), hippocampus (3-class), multimodal brain tumor (4-class), and nerve rootlets (9-class) via a modular factory-pattern architecture; loss-function ablations and foundation-model transfer learning cut labeling needs by 50%.
- Built a BIDS-compliant data pipeline over the Spine-Generic dataset and a Streamlit dashboard delivering MRI → segmentation → mesh → uncertainty in under 5 minutes per case (vs. weeks of manual CAD); authored a technical brief and PoC positioning the in-house AI stack against an external vendor for strategic ownership of CNS drug-delivery modeling.

Quantum-Enhanced Molecular Simulation & Retrosynthesis for Drug Discovery

Qiskit • PennyLane • VQE • ADAPT-VQE • DMET-VQE • CCSD • DFT

- Developed a VQE / ADAPT-VQE / DMET-VQE framework on 8–12 qubit systems achieving 97% correlation with CCSD/6-31G benchmarks for keto–enol tautomerism; reduced problem size from 184 electrons to 4–6 active orbitals while preserving chemical accuracy.
- Modeled HATU-coupling pharmaceutical synthesis with hybrid quantum–classical circuits benchmarked across multiple basis sets, cutting circuit depth up to 80% and variational parameters up to 60%.
- Applied DMET fragmentation (VQE for active regions with classical MP2/CCSD) to simulate 68+ atom drug-discovery-relevant molecules, achieving sub-kcal/mol accuracy on benchmark reactions.

EDUCATION

M.Tech, Artificial Intelligence • NIET, Greater Noida • 2022

B.Tech, Computer Science • GNIOT, Greater Noida • 2020